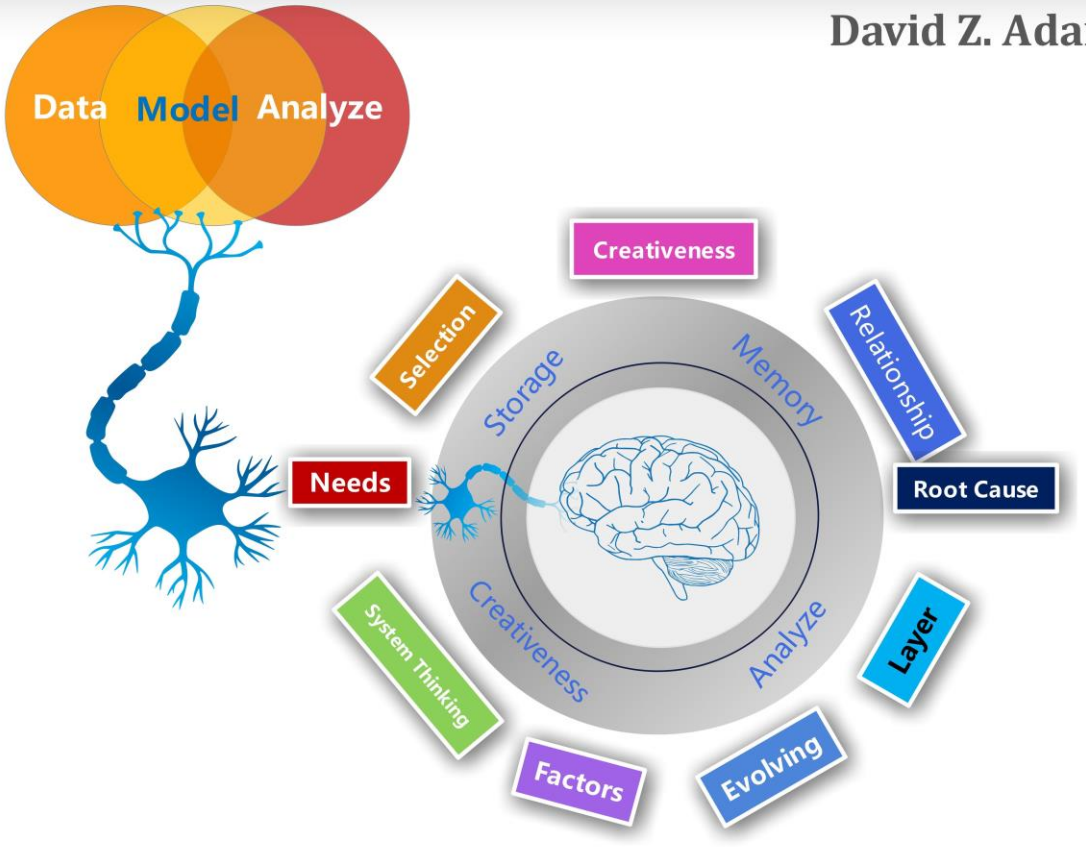
A Distillation of 5,000 Years of Wisdom

Actionable Thinking Models

A Revolutionary Thinking Tool for Achieving Multiple-Brain Power

David Z. Adams



Innovative Engine & Swiss Army Knife for Thinking

This Book Gives You more than,

12 DMA Thinking Models,

To Unlock the Power & Potential of Your Brain

We cannot solve a problem with the same level of thinking that we created it.

- Albert Einstein

The best investment is in the tools of one's own trade.

- Benjamin Franklin

The man who is in need of a new thinking tool, but hasn't yet acquired it, is already paying for it.

- Charlie Munger

You Need a New Powerful Thinking Tool in Your Life.

Now You Get it!

Preface

What makes a successful life?

Different people may have different answers, but undeniable that cognitive ability, problem-solving skills and creativity are essential for personal success, and all of these stem from strong thinking abilities.

Success comes in many forms, shaped by how we think and the scope of our ideas. Whether you're an entrepreneur, employee, scientist, businessperson, artist, or religious worker, strong thinking skills are essential for success in your field.

At the end of 2022, ChatGPT emerged, significantly boosting productivity, much like personal computers and the Internet once did. However, unlike other revolutionary tools, it quickly raised concerns about whether artificial intelligence would replace humans.

Even centuries ago, in a time without AI, people in social competition often feared their advantages would be replaced. A key strength of an individual is their thinking ability, which determines their future, especially during significant technological and productivity changes. A wise person can recognize this and make preparations in advance.

The book you are reading offers a comprehensive thinking method (A set of thinking models, named DMA Thinking Models) which will help you cope with various changes in the present and future, enhance your thinking abilities, outperform others, and achieve a fulfilling life.

Unlike the mechanical way AI extracts summaries from existing data, these thinking models leverage the advanced intelligence and creativity of the human brain. Readers can master them to build a strong foundation for their current and future competitiveness.

This book has undergone a five-year journey from conception to publication, crafted for those seeking to enhance their wisdom and thinking abilities. Readers will notice its uniqueness and find significant benefit from it. Before writing the book, I dedicated considerable time to gathering and analyzing works, speeches, and achievements of eminent thinkers, such as scientists and philosophers throughout history. I also studied the successes of practical achievers, like business executives, politicians, war commanders, and ordinary people who showed wisdom in overcoming difficulties. Additionally, insights on thinking and cognition theory and findings from modern scientists also help to understand how the human brain operates.

Based on those researches and studies, through comparison, extraction, integration, classification and summarization, a universal set of fundamental thinking method was formulated and is now ready for you.

To enhance understanding and practical use, this method is presented as 12 clear and tangible models, each with a vivid Visual Avatars to help you remember and apply easily. By engaging with these models, you will journey through time, learning the wisdom of both ancient and modern successful individuals. Since these models are from fundamental human thinking patterns that remain constant, ensuring their relevance in the future and providing lifelong benefits.

Imagine that behind this book lies the vast and illustrious team of super-intelligent people and experts. Their wisdom has guided humanity for millennia, becoming an inexhaustible source of human thought. Now, presented before you, is the condensed crystallization of ancient wisdom. We hope it acts as a guiding light, illuminating the path forward for you and inspiring the emergence of more wisdom and creativity for personal and societal development.

Notably, thinking models (or Mental Models) have been conceptualized for several years, readers can also find several books on this topic, such as *The Model Thinker* by Scott Page. These books showcase author's deep wisdom and abstract thinking, offering significant academic and theoretical value. However, this book takes a different approach. It is designed for a broad audience with diverse educational backgrounds, featuring models that are easy to understand and can be directly applied to daily work and life.

Each model is illustrated with examples of achievements from well-known personalities, celebrities, or successful companies, making them more relatable and convincing to readers. Additionally, the detailed examples used to demonstrate the creation of each model are drawn from situations that ordinary people commonly encounter in daily life. This approach ensures that both highly educated individuals and general public can quickly grasp these models and enjoy the learning process.

Many people have tried various techniques to improve their thinking in pursuit of a better life, only to eventually give up. They might attribute this to a lack of perseverance, but the real issue is often the complexity of the challenges they face and the ineffectiveness of the thinking methods they have. Despite having ambitions and ideals, they lack effective tools and give up halfway, which should not happen again.

Seeing things clearly, solving problems, and transforming the world are true sources of happiness. These align with the highest level of human self-realization. When people grasp the underlying rules of certain things or understand the reasons behind specific phenomena, they also experience happiness.

By choosing this book, you are taking a step toward truly learning how to think and solve various problems in work, study, and daily life while re-activating your brains via these visual thinking models. This will allow you to thrive with vitality, appreciate the world around you more deeply, and achieve lasting happiness and abundant wealth.

This book has a corresponding website (https://awiser.one) with a forum section that allows readers to interact, share their learning experiences, asking, answering or discussing questions. The forum will also feature the latest posts related to each model, encouraging users to analyze current events and hot topics through the lens of these models, gaining insights and happiness by observing the world and people from a higher and deeper level than others.

Elders often tell us to 'learn how to think' but no one has ever taught us how to think effectively and efficiently. For any problems, we mostly rely on personal comprehension or have to summarize from others with lots of efforts, time along with lessons and sufferings. Problems and challenges are endless, we need a useful and effective thinking method that can solve them.

This book is prepared for you who strongly wish to improve yourself with powerful thinking ability. It is never too late to find it, pick it up, learn and master it. You will only regret not discovering it earlier.

David Z. Adams

Seattle, Washington

Table of Contents

How to Use This Book i

Architectural & Relationship Diagram of Chapters ii

Chapter 1 What is DMA Thinking Model & How Can You Benefit from it? 1

Benefits-1: Enhance Workplace Competitiveness and Gain High Income 5

Benefit-2: Gain Inspiration From Innovative Thinking 6

Benefit-3: In-Depth Analysis, Solving Various Problems 9

Benefit-4: Efficient Learning & Understand Complex Knowledge Systems 10

Benefit-5: Facilitating Knowledge Discovery and Creating New Knowledge System. 11

Bonus-1: Reduce Mental Confusion & Anxiety, Alleviate Brain Fatigue 12

Bonus-2: Control Emotions & Respond Rationally to Unexpected Issues. 14

Chapter 2 DMA Model Overview & Tools Introduction 15

Section 2.1 Model Concept & DMA Model 18

Section 2.2 Introduction of DMA Model Creation Tool 20

Chapter 3 The Hub of Innovative Thinking - 9−Star Gathering Model 23

Section 3.1: Innovation Direction Pole 26

Section 3.2: Process for Creating Model Instance ..... **Error! Bookmark not defined.**

Chapter 4 Discover the Correlation between Things - Remote Control Model **Error! Bookmark not defined.**

Section 4.1: Process for Creating Model Instance ..... **Error! Bookmark not defined.**

Section 4.2: Model Upgrade - Quantification and Impact Analysis ..... **Error! Bookmark not defined.**

Section 4.3: Additional Applications of the Model ..... **Error! Bookmark not defined.**

Chapter 5 Identify Factors of Things - Building Blocks Model ..... **Error! Bookmark not defined.**

Section 5.1: Overview of the Factor Analysis Model ..... **Error! Bookmark not defined.**

Section 5.2: Prototype of Factor Analysis Model ..... **Error! Bookmark not defined.**

Section 5.3: Model Instance Creation Process, Identify Initial Factors ..... **Error! Bookmark not defined.**

Section 5.4: Model Instance Creation Process, Complete Steps ..... **Error! Bookmark not defined.**

Chapter 6 Things Development & Prediction - Ecosystem Model ..... **Error! Bookmark not defined.**

Section 6.1 Basic Components ..... **Error! Bookmark not defined.**

Section 6.2 Panorama of Ecosystem Model ..... **Error! Bookmark not defined.**

Section 6.3 Model Instance Creation Process ..... **Error! Bookmark not defined.**

Section 6.4 Tack Action & Develop Preventive Measures ..... **Error! Bookmark not defined.**

Chapter 7 Best Choice & Evaluation - Matrix Model ..... **Error! Bookmark not defined.**

Section 7.1 Model Framework and Components ..... **Error! Bookmark not defined.**

Section 7.2 Model Instance Creation Process ..... **Error! Bookmark not defined.**

Section 7.3 Dealing with the Factor preference - Model Disassembly ..... **Error! Bookmark not defined.**

Section 7.4 Two Special Considerations ..... **Error! Bookmark not defined.**

Chapter 8 Targeting Needs, Achieve More - The Barbell Model ..... **Error! Bookmark not defined.**
Section 8.1 Model Framework & Components..... **Error! Bookmark not defined.**
Section 8.2 Model Instance Creation Process ..... **Error! Bookmark not defined.**
Section 8.3 Widespread Application of the Model..... **Error! Bookmark not defined.**
Chapter 9 To Solve a Problem, First Understand it - Honeycomb Model**Error! Bookmark not defined.**
Section 9.1 The Object of Model Analysis - What is a Problem?..... **Error! Bookmark not defined.**
Section 9.2 Question Dissection - Honeycomb Model ..... **Error! Bookmark not defined.**
Section 9.3 Framework and Components of Honeycomb Model..... **Error! Bookmark not defined.**
Section 9.4 Model Instance Creation Process ..... **Error! Bookmark not defined.**
Chapter 10 Segment Complex Matters - Magnifying Glass Model ..... **Error! Bookmark not defined.**
Section 10.1 Model Components and Appearance..... **Error! Bookmark not defined.**
Section 10.2 Model Instance Creation Process ..... **Error! Bookmark not defined.**
Chapter 11 Identify Root Cause - Collapsible Telescope Model..... **Error! Bookmark not defined.**
Section 11.1 Model Components and Appearance..... **Error! Bookmark not defined.**
Section 11.2 Model Instance Creation Process - Problems Caused by Human Nature**Error! Bookmark not defined.**
Section 11.3 Model Instance Creation Process - Problems Caused by Natural or Objective Things... **Error! Bookmark not defined.**
Section 11.4 Proactively Using the Model to Achieve Personal Goals ..... **Error! Bookmark not defined.**
Chapter 12 Classification & Pattern - Filter Model..... **Error! Bookmark not defined.**
Section 12.1 Model Framework and Appearance ..... **Error! Bookmark not defined.**
Section 12.2 Model Instance-1 Creation Process - Forest Wildfire..... **Error! Bookmark not defined.**
Section 12.3 Model Instance-2 Creation Process - Johnson's Authenticity in Running for President .. **Error! Bookmark not defined.**
Section 12.4 Model Instance-3 - Predicting the behavior of presidential candidate McKinsey..... **Error! Bookmark not defined.**
Chapter 13 Four-Dimensional Decision Making - Coordinate System Model**Error! Bookmark not defined.**
Section 13.1 Model's Component and Framework ..... **Error! Bookmark not defined.**
Section 13.2 Right-linear Model → Advance with deduction ..... **Error! Bookmark not defined.**
Section 13.3 Left-linear model ← Goal regression..... **Error! Bookmark not defined.**
Section 13.4 Upper-linear model ↑ Grasping the big picture - future or macro perspective**Error! Bookmark not defined.**
Section 13.5 Lower-linear model ↓ Delving into details - historical or micro perspective**Error! Bookmark not defined.**
Chapter 14 Master Complex Information Systems - Neural Tree Network Model**Error! Bookmark not defined.**
Section 14.1 Model's Framework and Appearance..... **Error! Bookmark not defined.**
Section 14.2 Model Instance Creation Process - Understanding Professional Books**Error! Bookmark not defined.**
Section 14.3 Refining the Process ..... **Error! Bookmark not defined.**

Appendix I Brain Operating Mechanism - The Engine of Thinking ..... **Error! Bookmark not defined.**
Brain's 'Personality Traits' ..... **Error! Bookmark not defined.**
Brain's 'Machine Attributes' ..... **Error! Bookmark not defined.**
Appendix II: Relationships Between Things - Grabbers of Thinking ..... **Error! Bookmark not defined.**
Appendix III: Analysis Purpose - The Starting Point of Thinking ..... **Error! Bookmark not defined.**

How to Use This Book

Required Tools:

A pen and paper

Improving one's thinking ability is a continuous process that is honed through practice. This book offers more than just theoretical concepts; it presents a practical method. To fully grasp the methods introduced, readers could try to apply the models to their work, studies, and daily life frequently. Practices will yield the best results, and all you need is a pen and paper (preferably not smaller than B5 size) to draw any of the 12 DMA Thinking Models. You can also use an electronic tablet with a touch pen.

A computer & software

During the learning process, we don't recommend using a computer to draw model diagrams. The reason is that using a computer might limit oneself because of the inconvenience and lack of freedom in graphic drawing. However, for complicated models and future reference, you can use a software to create and store the any model diagrams.

Computers and software offer certain conveniences, but they are unnecessary for mastering the models. Ordinary pen and paper are more conducive to learning the models effectively.

If you prefer to use computer software for drawing models, it's best to use professional graphic design software like Microsoft Visio or those similar. While Microsoft Office's Word or Apple's Pages can draw basic graphics, you may need some time of learning and practice to create complex diagrams.

In Chapter 2's section 2, I will introduce some software for creating DMA models. I've tried all of them so know their pros and cons, hope you can choose one best for you. Again, the tools are not the most important part of learning and mastering DMA models, understanding and practicing frequently makes perfection.

Reading order:

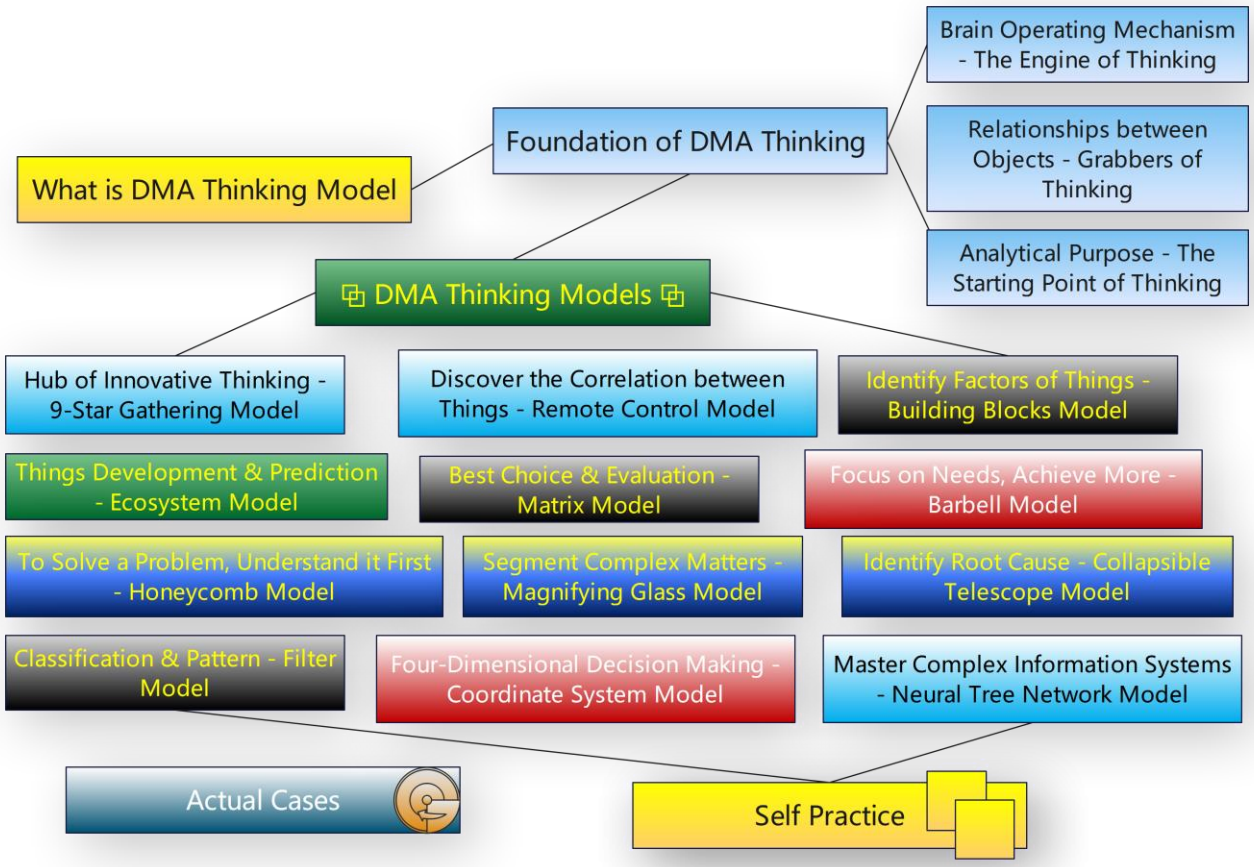
Except for the summary introduction of this thinking method at the beginning, readers will find each of the 12 models in its own dedicated chapter. There are some models are more fundamental, but if readers didn't read them first, that does not affect understanding of other models, or readers can easily go back to read any of them once being mentioned in another chapter.

The model names listed in the table of contents indicate one specific thinking scenarios each model can address, allowing readers to navigate among relevant chapters, making the book like a dictionary for quick reference.

Following model chapters, the appendices offer the foundation and principles for the creation and design of the DMA Thinking Models, which may interest readers who wish to gain a comprehensive understanding of the reasoning of how and why I design and made these models. This part is optional for reading, but I suggest you read it after completing reading the model chapters. It might inspire you with some new, interesting and creative ideas beyond the DMA models.

The following architectural diagram shows the arrangement and relationship of chapters in the book.

Architectural & Relationship Diagram of Chapters



Pair Learning Practice:

If you have friends who are also interested in thinking methods or seeking a wiser life, recommend this book to them and form a study group to learn, discuss and practice together. This not only makes learning more enjoyable and strengthens friendships but also leads to deeper and more well-rounded understanding and thinking.

You'll realize different people create different DMA models for the same analysis scenario (although the core parts should be similar). The variations in personal models reflect individuals' varying levels of understanding and perspectives, or their distinct objectives. Pair learning allows you to gain new and broader insights into problems and other people's viewpoints.

If you don't have offline friends to study with, you can also visit online forums and take part in discussions about any models and their examples. This expands your horizons, stimulates your mind, inspires new ideas, and also provides an opportunity to make new friends or find a good partner for your business or career development.

Chapter 1

What is DMA Thinking Model & How Can You Benefit from it?

Foremost, DMA Thinking Models are a set of 12 visual Thinking models, which can empower you in fulfilling your aspirations through improving your thinking ability and capacity.

Every rational person looks forward to a future that surpasses their current self - a life of greater prosperity, greater accomplishments, and more fulfilling relationships. The knowledge and skills of thinking abilities introduced in this book will serve as amplifiers and promoter in achieving your goals.

The 12 visual and easy to implement thinking models can be used individually or in combination, helping people to think effectively and efficiently, acting as one or more extended external brains for people from any education background and any field.

Each model is used to solve a type of thinking scenarios, such as using the **Selection & Evaluation Model (Matrix Model)** to make the best choice among multiple options; using the **Ecosystem Model** to see the future then plan your job promotion strategy.

And the same model can meet different needs. For example, the need to win business customers is the same as to keep a good marital relationship. We can achieve both by the **Needs-Oriented Model (Barbell Model)**.

How were DMA thinking models forged? By summarizing the common thinking processes behind the successful experiences of people from ancient Greek philosophers to today's successful entrepreneurs, inventors, scientists, politicians, military leaders, and other professionals, I distilled and designed these visual models. They reflect the fundamentals and commonalities of the most effective human thinking activities and can help learners enhance their thinking abilities, see the essence of things, and ultimately achieve career success, gain wealth, and attain harmony in relationships with others.

DMA is the abbreviation of the three words Data-Model-Analysis and is the three elements for a sound thinking process. **Data** is the basic information input required for a DMA Model. For example, when determining the best option among many options, all options and their attributes belong to the Data part. Sometimes Data is well known or default, otherwise the model will drive analysts to collect and supplement the missing Data parts.

Model refers to one applicable DMA thinking model, determines the input information (Data) it needs. **Analysis** is the process of attaching Data to the thinking model and then analyzing it as required and directed by the model.

Let me give an analogy. If we want to go to a distant place, we choose the model **M** - let's assume driving a car; then according to the model's requirement, we need to prepare gas and other necessities as the **D** parts serving **M**. Finally, **A** represents the action of driving.

Here's a specific model instance - using the Matrix Model for university selection and the Ecosystem Model for career advancement within a company.

Example 1: For a high school graduate choosing the most suitable university among several options, the **Data (D)** part comprises information about each university's location, reputation, ranking in majors, and career assistance, etc. The **Model (M)** part is the DMA Matrix Model itself. The **Analysis (A)** involves incorporating the Data into the model to complete the selection process.

Similarly, if you want to choose the best vacation destination among many locations, you can use the same model, but the Data part obviously differs.

Example 2: For seeking a promotion within a company, the Ecosystem Model can direct you to collect the requirements of the new position in terms of business and organizational skills; your strengths, strengths and weaknesses of competitors, and superiors' expectations. All these info form the **Data** part.

Then, the Ecosystem **Model** can use this Data to perform a deep and forward-looking **Analysis** of the competition process for the new position, and show you the best strategy and actions needed.

Below is a Table (Tab-1.0-1), displaying the categories, names, and application fields for each DMA thinking models. It may appear somewhat abstract now, but as read on, you will gain a better understanding for all of them. Below Tab-1.0-1, another Table (Tab-1.0-2) showcasing how specific thinking problems corresponding to a DMA model.

Both tables aim to provide readers with a comprehensive understanding of DMA Models. Readers could first identify their specific problem and then choose the corresponding model to address it[1] (Tab-1.0-2), or use models to inspire themselves with a new idea to resolve some unrecognized problems (Tab-1.0-1).

Models and Problems they Solve (Tab-1.0-1)

Category	DMA Model	Problem to be Solved	Visual Avatars
Basic	Factor Analysis Model	Identify the major factors that influence the current status and future development of things.	Remote Control
Advanced	Correlation Model	Uncover hidden relationships: • Relationships between components of things. • Relationships between external things. • Relationships between seemingly unrelated things.	Building Blocks
	Innovation Model	Solving problems with creative thinking and innovative approaches.	9−Star Gathering
Behavioral Guidance	Selection & Evaluation Model	Choose the best option among multiple options.	Matrix
	Needs-Oriented Model	Start with the needs of others and finally realize your own needs.	Barbell
Analyze Problems & Make Decisions	Four-Dimensional Decision-Making Model	Comprehensively understand problems from four dimensions, make rational decisions, and discover those overlooked things.	Coordinate System
	Problem Dissection Model	Comprehensively analyze and see the essence of the problem.	Honeycomb
	Problem Segmentation Model	Divide complex problems or things into multiple parts for better understanding and handling.	Magnifying Glass
	Root Cause Analysis Model	Find the root cause of a problem.	Collapsible Telescope

[1] Some problems contain multiple sub-problems, and different sub-problems may require different DMA Model(s) to solve. This naturally reflects our complex world. So it's important to understand and master each model to gain the most thinking power.

	Inductive Analysis Model	Recognizing patterns from common phenomena and see through the truth behind things.	Filter
Things Development	Things Development & Prediction Model	Based on the popular System Thinking, deduce and predict the development of things, and actively interfere it to gain benefits.	Ecosystem
Complex Information Systems	Information System Model	Use the information structuring and processing methods like human brain to comprehend and manage complex and extensive information & knowledge system.	Neural Tree Network

Below is the second table, which presents the relationship between different problems and the corresponding models with some analysis scenario examples, viewed in terms of the problem rather than the model.

Some problems may seem to be resolved by different models, that's because the problems themselves often encompass various facets. Analysts can first identify their most desired outcomes or the most crucial analytical objectives. Then, they can choose one most suitable model. If time permits, they can also employ multiple models to confirm results and gain more insights.

Sometimes using different models to analyze the same thing will yield different conclusions, but this is not usual and rarely causes confusion. Instead, it deepens the understanding of the problem from different angles.

Another situation is Model cooperation - means using multiple models to resolve one complicate problem. For example, to achieve new product designs with high sales prospects, use the Needs-Oriented model first to identify customer's requirement, then use the Innovation Model to design some fantastic features.

Problem and Models can use (Tab-1.0-2)

Problems and thinking scenarios	Model Matched
• What will be the effects if the US restricts chip supplies to China, and what are the potential long-term consequences? • How to overcome powerful competitors to win promotion in a company? • How to find opportunities from the development trends after implementing a new policy or measure? • What changes will this city have in the next few years or decades? How will they affect me, or how can I benefit from them?	Things Development & Predict Model (Ecosystem)
• How to discover potential investment opportunities in advance? • How to correlate current climate change to its effects on my business?	Correlation Model (Remote Control)
• How to understand and memorize a complex knowledge system (an academic discipline or a huge open-source software project)? • How to learn a new skill (foreign language, programming language, musical instrument, etc.)	Information System Model (Neural Tree Network)
• How to build lasting friendships with confidants? • How to maintain good superior-subordinate relationships? • How to win and keep customers?	Needs-Oriented Model (Barbell)

• How to achieve an annual income of 2 million? • How to choose a suitable career path? • How to avoid crowded or homogeneous competition when competing for resources with time or space constraints? • How to double my sales?	Four-Dimensional Decision-Making Model (Coordinate System)
• How to see through the purpose behind a politician's behavior or rhetoric? • With rumors flying, how to judge their authenticity?	Inductive Analysis Model (Filter)
• What undiscovered mineral deposits might exist in this region? • Does my ability match the requirements of my current job?	Factor Analysis Model (Building Blocks)
• How to choose the most suitable one from multiple brands and models when shopping? • Among two friends, how to determine the most suitable partner for me?	Selection & Evaluation Model (Matrix)
• Is the decline in sales performance this month due to my own actions or external factors? • The insufficient number of elevators causes excessively long wait times, What is the best way to handle it?	Problem Dissection Model (Honeycomb)
• This project is too large and needs to be divided into smaller parts to be handled separately. How can I divide it? • In the brainstorming meeting, everyone speaks freely, but we struggle to focus on the key issues that need decisions. How can we manage this?	Problem Segmentation Model (Magnifying Glass)
• In recent years, this area has experienced frequent torrential rains. What is the underlying reason behind this? • Many shining Chinese enterprises went bankrupt after initial success. What is the root cause?	Root Cause Analysis Model (Collapsible Telescope)
• None of the solutions are working. How can I use innovative thinking to solve the problem? • How can I design an innovative and eye-catching product?	Innovation Model (9−Star Gathering)

We face so many problems, most of them can correspond to a DMA thinking model to be analyzed and handled, or we can break down complex problems into smaller ones to be solved one by one using specific models.

You may already be interested in these models and are ready to start learning and using them? To boost your confidence and sustain long-term practice - which will require effort and time - I hope you read the following brief sections about the benefits of these models.

These sections are based on real-life cases. As you read, imagine yourself in similar situations, realizing your desires, overcoming challenges, and significantly improving your work and life in various aspects.

However, if you’re eager to jump straight into the specific model introductions, you can skip directly to Chapter 4, starting with the Innovation Model. You can always come back to read these Model Benefits sections to motivate yourself.

Benefits-1: Enhance Workplace Competitiveness and Gain High Income

Enhancing one's competitiveness in the workplace through improved thinking skills is crucial not only for employees but also for managers and entrepreneurs.

You and your colleague both aspire to be promoted, but there is only one promotion opportunity. How can you ensure you come out on top?

You know the ability to manage existing products and expand business through innovation can help give an edge over competitors, leading to increased profits and growth. But how to achieve these goals?

The table below just showcases some DMA Models used for some workplace scenarios. You can use more models for more situations after learning all 12 models.

Models Corresponding to Common Work Scenarios:

Work Scenarios	Adopt a Model - Brief Process
Complete daily work with high quality and high efficiency: • How to ensure Quality • Improving Time Efficiency • Make the Best Decision	**Factor Analysis Model**: Identify elements of quality improvement; analyze where we waste time. **Selection & Evaluation Model**: Based on the priority of different tasks, choose the most important one to be completed first. **Problem Dissection Model:** Analyzing the problem of inefficiency.
Carry out innovative methods to generate more value or reduce time costs: • Where to Begin Innovation • How to Judge Future Trends in an Industry or Technology	**Factor Analysis Model**: Identifying areas that need innovation. **Innovation Model**: Implement new work methods/process; develop innovative technologies and new products. **Things Development & Prediction Model**: Analysis of Pros and Cons, and Prospect of new company strategy. **Problem Dissection Model:** Find the current major issues, then decide to use innovative solutions.
Achieving a salary increase or promotion: • Assessing one's ability for Promotion and Salary Increase • How to create a sound and effective promotion strategy • How to master a complex technology	**Factor Analysis Model**: Identifying the factors that affect promotion and salary increase. **Things Development & Prediction Model:** Develop strategies and plans to compete with strong competitors. **Needs-Oriented Model**: Maximizing the likelihood of success in salary negotiation by meeting employer's requirements **Neural Tree Network Model:** Efficiently gain a comprehensive understanding of existing product documents and technologies.
Establishing Harmonious Interpersonal Relationships in the Workplace: • Keep and improve relationships with the boss and colleagues • Leverage positive interpersonal relationships to boost your performance	**Needs-Oriented Model**: Gain a thorough understanding of the needs of relevant teams, superiors, and colleagues, and provide value to others. **Factor Analysis Model**: Identify relationships that are beneficial for your personal development and maintain them effectively to ensure a smooth workflow.

The above scenarios are just a few representative examples of work scenarios. In reality, there are many more diverse situations that a suitable DMA model can address. Applying a model is much more efficient than personal brainstorming and can also inspire new ideas.

If solving a problem requires new knowledge or information that you do not have, the model will direct and drive you to gain it. Nowadays, there're many means to gain knowledge, such as search engines, chatbots, online or offline knowledge databases, etc., so know what to search is more important.

The World Economic Forum releases reports on ***The Future of Jobs*** every two years. Along with job trends, it also provides a trend report on the Top 10 Skills that will be most in demand in the future. Below are summary reports (Top 5 Skills) for 2018, 2020 and 2023.

Top 5 Critical Professional Skills Required:

2018 Report - Top 5 Skills	2020 Report - Top 5 Skills	2023 Report - Top 5 Skills
• Analytical thinking and innovation • Active learning and learning strategies. • Creativity, originality and initiative • Technology design and programming. • Critical thinking and analysis	• Analytical thinking and innovation • Active learning and learning strategies • Complex problem solving • Critical thinking and analysis • Creativity, originality and initiative	• Analytical thinking • Creative thinking • Resilience, flexibility and agility • Motivation and self-awareness • Curiosity and lifelong learning

It is clear to see from the reports that the most required skills are thinking abilities (analytical thinking or critical thinking). Sometimes, experts also ranked the ability to solve complex problems as one of the most required skills. Actually solving complex problems also hinges on thinking abilities. Very likely job experts will continue to rank thinking abilities in the top five for the coming years or decades.

Considering the era of ancient warfare, where physical and combat skills were highly valued, but it's commanders' wisdom played a key factor in determining victory.

The top five skills also often include innovation and learning abilities, which have their own unique requirements, but are closely tied to thinking abilities (there're 2 DMA thinking models for innovation and learning).

Benefit-2: Gain Inspiration From Innovative Thinking

We can observe that almost every seemingly invincible corporate giant such as Kodak and Nokia, has been defeated by companies with innovative products. The same applies to individuals - only through continuous innovation can one stay competitive and achieve sustained progress and success.

The following examples illustrate scenarios where we can apply the **DMA Innovation Model**. The **Process in Brief** column shows direction components of this model, outlining several directions for innovative thinking and provides a brief introduction (for detailed instruction, see Chapter 4). Its serial numbers match to those in column **Examples.**

Innovation Scenarios	Process in Brief	Examples
Develop new products	**1. Aggregation & Connection**: Integrating multiple functions into a single product or service. **2. Cross-Disciplinary Attributes**: Applying a certain attribute of something to a completely new field, expand product functions and fields of application. **3. Enlarge & Shrink**: Enlarge or shrink a product (or their components) to meet new requirements or expand into new areas.	1. Swiss Army Knife; Multifunctional Aggregation Chip. 2. Hydraulic Jack (principle of equal hydraulic pressure in the same plane, applied to mechanical engineering. i.e. Car jack) 3. Compressed biscuits; Projector.

Adopt new designs, new methods to transform existing products or services	**1. Cluster Expansion**: Increase the number of affordable products to combined into a collection, offering an alternative to a single expensive product, or achieve advanced functionality by combining different products. **2. Segmentation and Separation**: Divide a product or service into multiple components to simplify management, lower replacement costs, and reduce the risk of multifunction as a whole. **3. Cross-Disciplinary Attributes**: Transform a particular attribute of a product or service into another form to achieve new functionalities or solve certain issues.	1. Google employs a multi-server cluster instead of a single large powerful server. 2. Detachable ink cartridge fountain pen; Segmented computer subnets. 3. Fragranced baby diapers - Let the Liquid activate the scent to draw attention.
Employing innovative approaches and methodologies to address longstanding issues.	**1. Cross-Disciplinary Attributes**: Transform a blocking property or use properties of something else to solve a problem. **2. Expansion and Extension**: Extend or expand the original design in space and/or time to gain new functions.	1. Immersive Lithography Solution - leveraging water refraction to reduce light wavelength, overcoming lithography machine limits at a lower cost in chip manufacturing. 2. Slide rail for photography and videography - extends shooting angles.
Use new ideas to create harmonious interpersonal relationships	**1. Blockade & Isolation:** Moderate interpersonal conflict or reduce it by isolating the parties involved. **2. Aggregation & Connection**: Create opportunities to enhance communication and achieve objectives.	1. Separate individuals likely to conflict to minimize potential issues. 2. Invite talents that headhunters cannot recruit to participate in technical forums then connect them to our projects.

As shown in the examples above, the **DMA Innovation Model** is more systematic, practical in innovation practice, and easy to use. It can generate multiple innovative ideas for the same need, making the model engaging and enhancing the user's sense of accomplishment.

Almost all areas showcase innovation in various forms. While innovative ideas require divergent thinking, but unstructured or unfocused thinking can stray from goals. The DMA Innovation Model offers nine thinking directions (that's why it's called 9−Star Gathering Model) to guide and stimulate active, innovative thinking, making innovation more structured and goal-oriented.

You can find more details about these nine thinking directions in Chapter 4. Below is just a brief introduction.

Aggregation & Connection: Combine different components or **things** (entity, object, concept)[2] together.

Expansion and Extension: Extend or expand the functionality of existing things.

Segmentation & Separation: Separate or divide various components of a thing.

Blockade&Isolation: break the connection between different things or components of one thing.

Attribute Conversion: Transform certain properties of things into their opposite reverse form.

Cross-Disciplinary Attributes: Apply attributes to new domains.

Enlarge&Shrink: Significantly enlarge or reduce the volume or scope of the current thing.

[2] There are different types of things, physical entity, physical or abstract object, abstract concept etc. If "thing" used in the sentence, it means the model or its components work on all types of things, otherwise a specific noun will be used for clarity.

Cluster Expansion: Multiply the thing itself or a component to form a group, achieving functions that a single thing cannot provide.

Space & Time Reorganization: Rearrange the order of things or components in space and time to create new functions.

Yes, these introductions may seem quite abstract. Here are two examples: designing new products and coordinating interpersonal relationships, hope they can help you understand the nine thinking directions.

Design new products:

Imagine you are the owner of a syringe manufacturer in a saturated market. How can you use innovative thinking to design and produce novel products to expand sales?

Aggregation & Connection: When car owners go out for a long distance, they rarely carry a large window cleaner with a bucket. You can design a product that combines a medium-sized syringe and a brown brush - Fix the front end of the syringe with a brush, and when needed, spray water onto the brush to clean the windows and vents.

Segmentation and Separation: There are oil storage device that requires oil to be transferred through an external interface, for different devices the interface's diameter are usually different.

However, a common oil gun might not match that interface, design and produce a simple 'syringe' with just the tube body. Connect the input end to the oil gun outlet and equip the output end to the oil storage device. By preparing a set of tubes with various diameters, you can accommodate different oil storage devices.

Expansion and Extension + Enlarge & Shrink: Imagine a zoo where, when feeding animals, breeders must first lure and isolate the animal in another room, place the food, exit, and then open the adjoining door to let the animal return.

This process is time-consuming and tedious, especially since breeders must feed many animals several times a day. To simplify it, you can design a longer, thicker syringe as a food dispensing device. By removing the needle head and thickening the tube tip, this device would allow breeders to push food or water into the enclosure from outside, making the feeding process much easier.

Segmentation & Separation + Cluster Expansion: Before going on vacation, families often ask someone to water their indoor plants, which is inconvenient.

One solution is to create a large-scale injection needle with an extended tip that can be inserted into the soil of the flowerpot and filled with water before leaving. This needle will then slowly water the plant while the family is away. Additionally, you can use the Cluster Expansion direction by placing multiple needles with the same design in the pot if more water is needed.

Blockade & Isolation + Cross-Disciplinary Attributes: Imagine you have a fish tank with several beautiful tropical fish and want to add a small turtle while keeping them separated. To do this, you can use a thick syringe tube to hold the turtle and water, then place it in the tank, creating the illusion of the turtle and fish coexisting. This use of the syringe tube also falls under the Cross-Disciplinary Attributes direction, as it transitions from an injection tool to an isolation tool.

Some of the examples above may seem unrealistic, they are intended to familiarize you with these creative directions. In practical applications, there are nine directions for innovation, each potentially inspiring multiple new ideas. Choose the most appropriate direction(s) based on the actual situation.

Coordinating interpersonal relationships in a company:

Aggregation & Connection: If there are many disputes, inefficiencies, and obstructions between two departments, consider merging them into one.

Expansion & Extension: If a department cannot obtain the necessary inputs directly and cannot resolve the issue by collaborating with other departments, it may be beneficial to establish a direct connection between the department and the resources. For example, if a company's research and development department is highly motivated but lacks the ability to engage with customers for prompt product feedback, the company may create a direct link between the R&D department and the customers.

Enlarge & Shrink: If there are issues with costs or efficiency, the department can be expanded or downsized. Downsizing, such as reducing redundant staff, is straightforward. For expansion, consider a situation where department staff frequently travel to a branch office, incurring high travel and coordination costs. In such cases, expanding the branch offices to handle some functions independently can be a beneficial solution.

Attribute Conversion: If budget constraints prevent expanding the customer service department, you can have the Training Department collect customer feedback on products and new feature demands. This approach shifts Training Department from being information provider to receiver.

Benefit-3: In-Depth Analysis, Solving Various Problems

Usually, a problem triggers people to think. Each DMA model corresponds to a specific type of problem.

Before diving in solving a problem, we need to understand the problem well to avoid deviation caused by misunderstanding. Some DMA Models are designed for understanding a problem. These models can enable users to quickly comprehend the problem and find a solution. In over 50% of the cases, when people truly understand a problem, they can easily find a solution.

There are 2 models for problem understanding: **Honeycomb Model**, **Magnifying Glass Model**, and **Collapsible Telescope** Model for finding problem's root cause:

The sales performance has decreased this month. Where did I go wrong?	Honeycomb Model
Why am I getting trapped in social media every day and wasting a lot of time?	Magnifying Glass Model
In recent years, many some companies have gone bankrupt. What is the reason behind it?	Collapsible Telescope Model

Secondly, there is another big category of problems: How to select the best among multiple options:

Among several universities with distinct features, advantages and disadvantages, which one should I apply?
I have received offers from two companies, while salary is just one factor to consider, I am unsure which one to choose.
Which major is more suitable for me: Computer Science or Finance?

Apply the **Selection & Evaluation Model (Matrix Model)**, we can solve the selection problem quickly and efficiently.

As a member of society, many of the problems we face originate from the exchange of interests with others (or conflicts of interest). If we do not solve this first, even a sound solution may fail.

For example, when designing a new product, if we only focus on using the Innovation Model to create a truly unique product, users may not like it. This raises another issue: how do we understand and uncover the actual needs of others? Similar challenges are:

What strategies should I use to persuade my boss to give me a salary increase?

What type of gift should I buy for my wife that will surely make her happy?
My YouTube channel has a few subscribers. How can I improve?

For this kind of problem, the **Needs-Oriented Model** helps. With this model, the selection process will be much more efficient and ease your mind.

The 4th type of problems is how to judge or predict things development trend or how to make a strategy which will be impacted by many other things and need a long period to be realized.

What steps should I take to plan for a promotion to a higher-level position?
How can I achieve an annual income of $1 Million in five years?

Although the goal seems simple and clear, but it involves many factors which cause uncertainties. For this type of challenges, we can use **Things Development & Prediction Model (Ecosystem Model)** to make a sound analysis & action plan.

Another common challenge is we need to judge things and people from appearance, such as:

Former President Trump mentioned in public that the Democratic Party had fraudulent behavior in the election. Is he justifying his failure or is it true?
In recent years, wildfires have frequently erupted in California, and similar situations have occurred in other places of the world. Through comparison, we can assess that thick forests dominate these areas, coupled with high temperatures. Is this judgment reliable?

For this type of problem, the **Inductive Analysis Model** (Filter Model) is a powerful tool to help us gain a deep insight.

The examples mentioned above are only a partial representation. The 12 DMA models provide comprehensive coverage of various problems and challenges that we may encounter every day, like career development, mastering new knowledge, gaining innovative ideas and interpersonal interactions. By mastering these models, you can gain deep insight and understand much more deeply than others, also the appropriate model(s) can help you handle them effectively.

Benefit-4: Efficient Learning & Understand Complex Knowledge Systems

This is an era where new knowledge and technologies are emerging in an endless stream. The good thing is, with the willingness to learn, one can always find suitable resources and materials. However, this also brings pressure and anxiety, as the complexity and diversity of knowledge requires significant time and energy to understand.

When knowledge from various fields floods in, it becomes even more challenging for us to learn and master. Although we can stand on the shoulders of predecessors to gain their knowledge, the sheer volume and complexity of information can be overwhelming.

Is there a shortcut to help us master complex knowledge?

First, let's see what the most complex things look like. Scientists have published a paper revealing that the structure of the human brain and the universe are almost identical[3]. Both the universe and human brain

[3] The paper, published in 2020 by University of California, Los Angeles researchers Franco Vazza and Alberto Feletti, found striking similarities between the human brain and the makeup of the universe.

are composed of huge interconnected parts, much like a tree that grows from a seed into two buds, and the buds develop into trunks which develop into branches, then leaves.

The DMA **Neural Tree Network Model**[4] was designed based on this finding and is specifically for learning and understanding complex info-system, such as open-source software systems, academic disciplines, or any other complex things, enabling people to grasp it as a whole along with the details in-depth.

Even a complex knowledge system is far less complicated than the human brain. By applying the structure of the brain to analyze and break down a particular subject, its complexity becomes insignificant. The internal composition of correlation and interaction can also be well observed and easily understood, making it more accessible for us to comprehend and master.

Let's take the example of learning a complex subject by reading a professional book (In Chapter 14, you can find the detailed model creation steps). The presence of a book's table of contents serves as the main framework (corresponding to the major chapters and sections). When building a Neural Tree Network Model, first draw these main branches, and then quickly extract the summary of each chapter and add it to the corresponding branch in the model. This step helps us with a basic understanding of the subject logical framework with minimal time investment.

Next, extract the summaries of each section under their respective chapters and add them as a third level branch. At this point, you will have a foundational understanding of the entire book.

In the Chapter 14, you will also learn a technique where you could imagine yourself is the original author, which spurs you to flesh out the book's specifics after comprehending its general framework. This will deepen understanding of the author's thinking process and aids in memorizing.

When dealing with a system that does not have a pre-existing trunk and branch structure, such like a huge open-source software, creating a Neural Tree Network Model from scratch can be outlined as follows:

Starting from the main function or the home page file (index.html, etc.), according to the division of internal logic, and corresponding to each source code file, sort out each module. Then list the components of each module, break each part down into singular class, function, etc.

One aspect that distinguishes Neural Tree Network Model from an actual tree is that there are additional direct or indirect relationships between the same or different level branches of the model, which can be represented through connecting lines. Various relationships, such as Dependency or Correlation, can be depicted using different line colors and styles, resulting in clear and intuitive visualization.

Benefit-5: Facilitating Knowledge Discovery and Creating New Knowledge System.

Perhaps you heard of some software companies in China implementing a 996 work schedule (extending daily working hours from 9AM-9PM, 6 days a week) where employees earn significantly more than other companies. Is this work schedule better than the traditional 8-hour 5-workday?

[4] The **Neural Tree Model** reduces number of branches, retains the essential structure, and it looks very similar to the existed Mindmap thinking model, but the creation process of the two is different. The Mindmap is that the creator first has the branches in his heart and then add the details freely, while each component of the neuron model comes from the Thing to Be Analyzed, which is purely objective. Since the brain structure and the essence of the thinking process are the same, the two thinking models are similar in appearance.

Li Yongle, a Chinese Internet celebrity teacher, has a video analyzing whether the 996 working style is beneficial to employees. He cited the economic knowledge that increasing marginal costs lead to diminishing marginal utility and concluded that the excess income got by 996 jobs is getting lower and lower, but the working hours are relatively longer and longer, obviously unprofitable.

Not everyone has a background in economics. If we don't have the knowledge of marginal cost, how should we analyze these problems? In such cases, we are completely ignorant of what knowledge is needed.

This situation differs from the complex knowledge system mentioned in the previous section that we can solve by learning it first. Similarly, some work-related problems involve already existed knowledge or skills that we were unaware of, or certainly they don't exist. When these hinder our progress, how to solve them?

For readers who do not have knowledge in economics, but are familiar with DMA models, they may opt for the application of the **Selection & Evaluation Model** (Matrix Model) to analyze the problem: whether to accept the 996 work schedule. There are two options: accept or not accept, and various factor items with weights will be used to ultimately decide the conclusion. The analysis process will necessitate the inclusion of wages and working hours as factor items[5], and the analyst can realize the connection between the two factors.

When one factor changes, both need to be analyzed together to effectively judge. For instance, if overtime exceeds 1 hour, the wage must be increased at the same or higher proportion to justify accepting it. If the proportion is lower than this, it should not be accepted - this embodies naive marginal cost knowledge.

For readers who have never learned about it, the application of naive marginal cost knowledge through models is enough for analyzing this problem. Those who like to delve deeper and expand their knowledge may extend it to create a knowledge system like marginal cost and more - this process is how knowledge develops from practice. In natural sciences, this intrinsic driving force is also strong and serves as a catalyst for many new ideas and knowledge to emerge.

In summary, the DMA model's requirement for input part of Data (knowledge or common sense, information, etc.) helps us identify the deficiencies in current Data and also prompt us from which angle or where to find and supplement it. Besides the Selection & Evaluation Model (Matrix Model), all other DMA Models can enable us same way.

This acts as an internal driving force, enabling us to ultimately solve one current problem and those related. And based on this, some more astute individuals may create a new knowledge or method/skill systems, extending it from a specific problem to wider application domains.

Bonus-1: Reduce Mental Confusion & Anxiety, Alleviate Brain Fatigue

Besides the benefits mentioned earlier for work, business, learning, and interpersonal relationship, using the DMA model can also help reduce anxiety and overthinking, allowing the brain to stay more focused during the thinking process, as well as prevent harmful thinking patterns from affecting both the brain and emotions.

[5] A Factor item in the **Matrix Model** (as the Data part) is a category used to help judge one aspect of the choices, for example, to choose the most suitable one for relocation among many cities (choices), climate is a factor item. For a specific model, there will be multiple appropriate factor items.

According to the latest findings in neuroscience, the brain's primary energy consumption occurs in the Default Mode Network (DMN), which accounts for 60%-80% of the total energy consumption.

Many people experience indecisiveness, even in simple situations, such as choosing to buy a USB flash drive, where factors such as capacity, speed, price, brand and comments can lead to long time comparing but still not sure which one is the most suitable.

How about significant life choices, like selecting a university or a career path?

People often struggle with issues that involve multiple points of consideration and multiple pathways of thought. This is because these issues involve various components and relationships that create numerous possible combinations of thinking paths.

For those who have not used the DMA Thinking Models, they will naturally use an innate and direct way of thinking, which combines all thinking elements (input, output, processing) in the brain, lack of systematization and hierarchy, the thought paths established between multiple relationships often change or difficult to preserve in memory. This can lead to prolonged periods of anxiety and difficulty in finding optimal solutions, even for simple problems.

The DMA Models encompasses various areas that need effective and efficient thinking. It provides an actionable and rational approach, guiding analysts to focus on both thinking elements and thinking processes and yield reliable and effective results within a reasonable time frame. Also, the model's repeatability and objectivity gains users trust to use them more and more over time[6].

As an example of the decision-making for purchasing a flash drive, the **Selection & Evaluation Model (Matrix Model)** can easily resolve it. For all Data parts, such as capacity, memory type, speed, price, stability, brand, and their respective quantified importance values, can be listed as factor items. Then we'll calculate scores for each flash drive option, finally select the one with the highest score.

Without using the model, one may waver between wanting a high-speed drive and not wanting to pay a high price, or choosing a compromise option and worrying about the stability of the memory chips. This can lead to mental fatigue with wasted time, and potentially ended up selecting the one not the most suitable.

In more complex situations, such as standing out among multiple competitors for a promotion, there are more thinking components and combinations of factors as of thinking paths. These may include comparing one's abilities and past achievements with others, superiors support, requirements of the new position, potential strategies that competitors may adopt etc.

With the support of the DMA **Things Development & Prediction Model (Ecosystem Model)**, all these elements and paths can be visualized and analyzed, easing the brain's memory load and minimizing repetitive thinking. It also enables prompt identification of any missing or insufficient aspects or elements.

What's more worth mentioning is, driven by the model(s), analysts get inspired to expand their thinking and take various measures to expand their probability of success, this makes the practice more interesting and exciting. Without the model? It might be continuous sleepless nights and restlessness.

[6] It should be noted that even using a DMA model, suboptimal result may be obtained, often due to missing key Data or Data is inaccurate while model creators/users are usually aware of this. With the rigorous and objective model thinking process, they will not lose trust in the model although it fails to meet their higher expectations.

Bonus-2: Control Emotions & Respond Rationally to Unexpected Issues.

It's quite common to encounter individuals who struggle to manage their emotions effectively. Many people are intelligent and handle various situations with clarity and astuteness. However, when facing unexpected issues, they often let their emotions take over, leading to significant losses and damage to their personal reputation. Is this a matter of lacking thinking ability?

Obviously not. If the problem happened to someone else, they usually demonstrate enough analytical and judgment skills to offer good solutions. So why do they lose their rational thinking when it comes to themselves? The reason is simple: they become too emotional when facing sudden problems, such as anger, frustration, feeling offended, or rushing to retaliate. These subjective emotions cause them to lose their sanity and make regrettable decisions based solely on intuition.

This is another bonus of DMA model thinking: when feeling emotional about a situation, quickly draft a **Selection & Evaluation Model** (Matrix Model) on paper, listing different ways to respond and conducting a simple analysis to predict the outcomes of each reaction. This approach can help calm oneself down and reveal how ineffective emotional responses can be[7].

Next, quickly create another model, such as a **Things Development & Prediction Model** (Ecosystem Model), a **Correlation Model** (Remote Control Model), or an **Innovation Model** (9−Star Gathering Model) to generate more ideas and insights. These models can help you think more clearly, inspire new ideas, and lead to a better response or action.

The DMA model's approach to managing emotional stress responses is clearly different from the methods suggested by psychologists or Emotional Quotient experts. It doesn't focus on suppressing or redirecting emotions but instead uses a structured model to quickly regain rationality, finding a better solution and allowing one to recognize the drawbacks of emotional reactions.

For instance, if your boss is criticizing your report after he lost a major business deal, you may feel frustrated and tempted to defend yourself. Using the **Problem Dissection Model** (Honeycomb Model) can help you find the cause of the issue: your boss's dissatisfaction is not related to your report, but the loss of an important contract. Therefore, the focus should be on finding a solution rather than fueling your own anger with your boss's.

The best approach is to handle the situation calmly and provide recommendations or solutions (possibly using the **Innovation Model**) in your report. This will demonstrate both your loyalty and your capabilities, turning the initially unfavorable emotional situation into an opportunity for career development.

Another suggestion is to prepare a table of different categorized scenarios and corresponding models for addressing different emotionally charged issues in advance. When a problem arises, refer to the table (in your phone, notebook or your brain) calmly and select the appropriate model for a quick resolution.

[7] Roman Emperor Marcus Aurelius, a key figure in Stoic philosophy, used a similar method to quickly calm emotions and make rational choices. However, this is not the source of inspiration for the design of the Selection & Evaluation Model; It only confirms that the practice of this model has existed since ancient times.

Chapter 2

DMA Model Overview & Tools Introduction

After a jet to the benefits of the DMA models, this chapter will summarize all 12 DMA models. These models summarize and distill the thought processes of human elites spanning history and territory, reflecting the coordination of the development and change of things and the way of thinking of the human brain. They can help users analyze infinite, objectively changing people and things, and solve various problems and challenges.

Based on the visual characteristics of the models, readers will come across various visual elements, such as charts, diagrams, tables/forms, and some other type of illustration. Each model chapter begins with several real-world examples to avoid abstract concepts that may cause difficulty in understanding. It is important for readers to truly comprehend the models, understand their design purposes and usage, and not be content with just learning a few examples. The forum module on the accompanying website (https://awiser.one) can help readers to take part in learning and exchanging ideas with others, broadening their horizons and perspectives.

Each model has a avatars name with corresponding visual image (such as matrix, ecosystem, barbell, etc.), making it easier for readers to remember them. Someone may think that this is not be so cool and like a book for young people. Believe me, you will appreciate the benefits of being able to recall and apply the models when needed with their avatars name and images. Imagine you are presented with ten people's names on a screen. Even you possess a natural talent, can memorize them on the spot, would you be able to know their personalities? Would you still remember them after a month? But, if we add ten photos corresponding to each name, the effect would be significantly different.

Although the DMA models may appear interesting, easy to understand and use, it is not designed for young audiences[8]. Readers should have at least a high school education level to truly understand its essence and design purpose to create their own effective models. The model is also not designed for specific professionals and does not involve industry jargon or complex calculations. Instead, they are a set of basic models suitable for the public. The model itself does not involve knowledge of monitors and sensors, but it can offer an effective foundation for innovation and help designers with this knowledge to expand their ideas to create appealing products. It can also assist a salesperson in analyzing sales difficulties and improving sales, or aid a college graduate in selecting the most suitable career field.

Why are these models created and what is the basis for their creation? The table below outlines the reasons for creating models - the fundamental purpose is to help model users better leverage their brain potential and increase their own benefits. The basis for creation these models is the decisive role and influence of the respective laws of the human brain and things around, as well as their interaction reflected upon human thinking process. (Please refer to Appendices I: Brain Operating Mechanism - The Engine of Thinking).

There is a popular term nowadays known as the '**second brain**' which refers to modern digital technology helping store and enhance memory functions of the brain. It is like adding a second hard drive

[8] To enhance children's thinking abilities, parents familiar with the DMA models can engage their children in fun and accessible model designs, instilling positive thinking habits from an early age.

to a computer to expand its capacity. Although the design purposes of the DMA models do not include the concept of second brain, the implementation process of all models involves systematizing and visualizing the majority of thinking processes in the brain through paper/pen or computer screen. This can be considered as an extension of the brain's functions, where the models not only serve as temporary or long-term storage devices but also get the computational functions of the brain, like adding one or multiple hard drives, memory sticks, and CPUs to a computer at the same time. For details, please refer to section below.

The 12 DMA models cover almost all scenarios that need thinking abilities. Users can refer to the summary content in the following table and the two example-filled tables in the chapter 1, looking up and apply an appropriate model to various aspects of work and life, become a wiser individual and achieve a fulfilling life.

Model Creation Reasons & Basis

Chpt	Model	Creation Reason(s)
4	**Factor Analysis Model**	Things are complex and have many components. In order to understand things and then analyze and grasp the laws of existence and development, identifying and understanding of thing's important constituent parts is an indispensable first step. This will enable us to view things holistically and provide us with diverse perspectives for problem-solving and innovative inspiration.
5	**Correlation Model**	There are multiple connections between things, and discovering or unearthing those that are not obvious or concealed can help us gain a first-mover advantage. For existing knowledge or exploring unknown areas, relationships can help us better comprehend and retain knowledge.
3	**Innovation Model**	The driving force behind social and economic progress comes from people's desire for a better life, and innovation is the fastest way to achieve those needs. It provides advantages over competitors, improves productivity, creates new tools, and avoids excessive competition. Innovation is also challenging, having logical thinking skills alone is not enough. We need a powerful and effective tool to facilitate innovation.
7	**Selection & Evaluation Model**	The reason we are what we are today is that of a series of choices made over the past few decades: choosing to study music or opting for a conventional primary school, deciding whether to attend college, selecting a major, choosing a company, and deciding a life partner. Even the daily decisions, such as whether to buy a house or rent, which apartment to choose, or which brand of phone to buy, can affect one's life trajectory and quality of life. A rational, powerful, and efficient tool for making choices that best represent our interests can prove immensely helpful.
8	**Needs-Oriented Model**	People, born for themselves, naturally put their own interests and needs first. To maximize benefits, we must first consider meeting the needs of others, so that others will buy our products, listen to us, considering our requirements, so that we can finally get what we want.
13	**Four-Dimensional Decision-Making Model**	Decision-making is like making choice, but more complicated, such as the comparison between 4D and 2D images. In order to make correct decisions more scientifically without regret, the Four-Dimensional

		Decision-Making Model (Coordinate System Model) provides 8 directions of thinking paths. Comprehensive decision-making process ensures the decision quality.
9	**Problem Dissection Model**	When facing a problem, people often start from the initial feeling. They usually haven't fully understood the problem yet, and their analysis and solutions based on intuition can easily go astray and become ineffective or fail. The Problem Dissection Model is used to gain a clear understanding of the problem, quickly solving most problems at the beginning or helping to confirm the correctness of the strategy and solution.
10	**Problem Segmentation Model**	For a single and complex thing or problem, before considering the relationship with other things, split it into multiple components (naturally, logic-based or purpose-oriented) is an effective means of analysis. Problem Segmentation Model can help us achieve optimal divisions.
11	**Root Cause Analysis Model**	After fully understanding a problem, in order to thoroughly solve it, necessary to find out the root cause. This can be a challenging process because of the complexity of the problem itself and interrelationships with internal and external factors, as well as potential hidden connections with other things. Effective tools are essential for this, as the Root Cause Analysis Model will help us discover the root cause of the problem from both human nature and the objective world.
12	**Inductive Analysis Model**	How to see the essence through the phenomenon? How can we find broader patterns that exist within ordinary things? These challenges are not only necessary for scientific research but also crucial for everyday work and social interactions. The Inductive Analysis Model, based on Mill's Methods[9], incorporates visual elements as of tables to enhance the clarity and convenience from original textual analysis process.
6	**Things Development & Prediction Model**	The development of things is not isolated; things tend to be chaotic and disorderly in their own development. To protect our own interests and decrease or avoid potential losses, we need a systematic thinking method to analyze and handle the set of all related things, people and their development and changes. Furthermore, we may need to adapt or change the development path of existed trend. The Things Development & Prediction Model can help us achieve all these objectives.
12	**Information System Model**	How to quickly comprehend and memorize complex knowledge or any system that is voluminous and intricately related? This challenge has been with us from childhood to middle age, and as our desire for success increases, so does the intensity and quantity of this challenge. Human brain's capacity is entirely capable of overcoming and conquering these challenges, while brain itself can be strengthened and enhanced. We just need a powerful and effective method to use our brain. The Neural Tree Network Model is designed for this purpose.

[9] Mill's Methods are five methods of induction described by philosopher John Stuart Mill in his 1843 book *A System of Logic*.

Section 2.1 Model Concept & DMA Model

Before diving into the details of each DMA Model, let's briefly define what we mean by a model (including certain simulators). The term 'model' refers to a smaller object (in physical, paper, or other form) created by people to simulate another complex thing, with a certain external visual or abstract expression, that can reflect or embody the external characteristics, internal mechanisms, and development trends of that complex thing. In the real world, we can see three types of models.

1. Simply imitate something in appearance

This is the most common type of model, such as a Boeing 777 plane model, a miniature representation of the actual plane. A plastic atomic nucleus model mimics the actual structure of the nucleus. These models simulate real and tangible objects but cannot have any substantial impact on the original objects. However, they can help people understand and appreciate the appearance, structure, and some basic operational mechanisms of the original objects.

There are also many functional models, such as a car engine model that can be powered on to mimic a real engine. It uses a brownish-yellow water to simulate gasoline and water steam to simulate oil vapors. The transparent outer shell allows for clear observation of how gasoline flows inside the engine and how piston movements are achieved. The purpose of creating such a model is not only for visual presentation but also to empower people to gain a deeper understanding of the operating principles of things.

2. Simulate the function or action of things through models

If you have seen the movie 'Titanic', you may remember the opening scene of underwater exploration of the sunken ship. The exploration team placed a probe into the seabed and then used special glasses and gloves to manipulate the probe's movements from shore, allowing its mechanical arm to do the same actions as if it were on site. This is a type of model which has no similarity with the original thing in appearance but can replicate some of its functions and characteristics. For such models, 'simulator' is a more accurate name. In some mobile app development processes, engineers also use mobile simulators to operate on their computers.

Economists also create various models to simulate the operation of social economics. These models use multiple levels and parameters to accurately analyze the current state of economic operation, predict future developments, and simulate the impact of certain external economic activities or events on current status. These models are used to represent, understand, or manipulate larger, abstract things.

3. Facilitate the work of greater things

Models of this kind are rarely seen in the real world, with the comparable examples being the writings and teachings of certain philosophers and religious prophets. These writings and teachings serve as models that offer profound insights into external realities and human nature, and influence those who read and understand them to apply the principles of the model to their own lives, creating a more orderly and harmonious world.

Overall, a DMA model is a simulation of the abstract thought process of the human brain, while also promoting the brain's capabilities and abilities in-depth and breadth during usage. As a result, all DMA models have characteristics of both the second and the third type of models mentioned above. They can help users in aligning with the brain's operational mode and making full use of brain function, while it's also

driving and controlling the brain to think and operate in the direction and manner set by the model and model creator/user. The abstract thought process is hard to understand through language alone, so all DMA Models are presented in a visual form to make it more accessible.

DMA models also have the characteristics of the first type of model. For example, the Barbell Model (Needs-Oriented Model) uses left and right connections and lines to simulate the correspondence between action and demand. There are also some models, such as the Ecosystem Model, like the second type of model, which simulates the operation and prediction of things while users can gain ideas to change things' trend via the model.

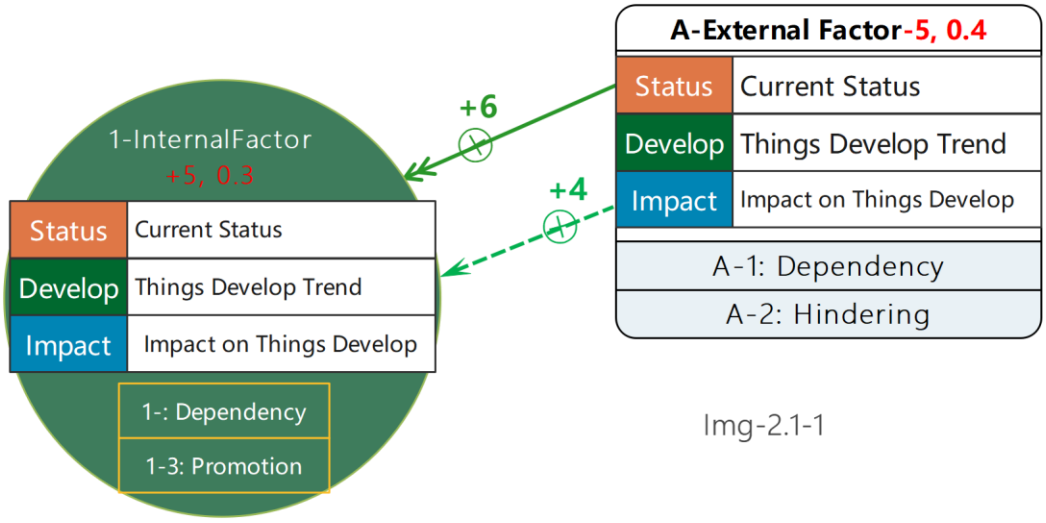
We know that the human brain is still far from being fully developed and effectively used. The conventional way of using the brain is like driving a Ferrari in a 25-mph speed limit community. By learning and mastering the DMA thinking models, readers will be able to truly drive the race car on the track and take it a step further by optimizing and upgrading the track as well.

Besides the above three types of models, the DMA model also realizes the function of the brain to store information. Any DMA model moves the contents of the brain to a visual model that exists on paper or in computer files (refer to the Brain Operation Mechanism in Appendix I), allowing the brain to free up space and time for more effective analysis, thinking, and innovation functions. The DMA models can now act as external brains, extending to visible and controllable objects and becoming a functional model of the brain.

Since the concept and process of 'brain thinking' are incredibly complex and extensive, no artificially created 'perfect' model can fully simulate it. Even if such a model existed, it would be quite challenging to comprehend and use because of its complexity. Thus, it employs multiple DMA models for different brain thinking patterns and processes in various scenarios. These models are categorized based on the ways of problem analysis and resolution, applying to almost all fields that need thinking and innovation. And these models are fundamental, so readers can create and develop their own specialized application models to improve themselves and benefit society.

As discussed before, the DMA model serves three functions: simulating the brain, guiding/driving the brain, and extending the brain. So, how does a particular DMA model achieve these three functions? Let's first look at the basic structure of the model. All DMA models comprise components, which are visual parts of the model. These components are classified into four static types: **shape**, **relationship**, **value**, and **assistant symbol**. The visual appearance of these components may vary depending on the model, but their usage remains the same.

Taking the example of a part of the Things Development & Prediction Model (Ecosystem Model), let's examine the components in detail (see the Img-2.1-1). A green circle denotes an internal factor of the entity under analysis, which is categorized as a '**shape**' component. Another 'shape' component, which represents an external factor of the same entity in a white rectangle. Between the two shapes, there are varied styles of lines representing '**relationship**' components (2 in the demo diagram), connecting and signifying the relationships between shape components.



Img-2.1-1

The numerical values adjacent to these relationship lines show lines' (relationship's) strength or intensity, categorized as '**value**' components. And the model employs some '**assistant symbol**' such as the ⊕ and ⊖ symbols attached to the relationship lines.

A model composed of static components can only express the state of things at a certain time. The dynamics of a model, however, refer to the interaction between the creator/user and the model, which is the practical comprehensive thinking process of the DMA Model method. Details related to these aspects will be discussed in the chapters of each model.

Before introducing each model, please note the following two points.

First, the creation and use of any model (including design, components, external forms, dynamic ideas and interactions, etc.) is based on the purpose of analysis and also serves the purpose. When using models, always keep this key point in mind: Creating models is not for fun or to showcase abilities, but to achieve the initial purpose(s).

Second, some readers who are familiar with Mind Map[10] may find that a few DMA models have a similar external form. This feeling is true because the brain's thinking process itself is multi-branching and layered, and any form of thinking process cannot avoid this structure.

However, unlike Mind Map which mainly serve as a means for the brain to store and simply extend branches, DMA Models achieve various dynamic thinking in different scenarios through different visual images and some calculation tables, and the determination of branches is no longer based on subjective feelings, but on specific purposes. And there are usually different relationship connections and quantitative values between different hierarchical branches/components - this more comprehensively simulates and promotes the brain's thinking process.

Although some DMA Models still adhere to the pattern of hierarchical thinking in the brain, their visual appearance is distinctly different. And the design principles and implementation processes of these models vary significantly depending on the analytical scenarios.

Section 2.2 Introduction of DMA Model Creation Tool

In China, there is a 2400-year-old saying, 'If you want to do a good job, you must first sharpen your tools.' In the West, a similar saying, 'You are only as good as your tools.' This principle applies to the creation, use, and development of DMA Models.

The tools can be as simple as a sheet of paper and a pen[11] or any computer drawing software. Though software is more convenient for modifying and long-term preservation, the traditional paper-and-pen method can still be useful, especially in situations where computer use is not possible or in a scenario requires quick model creation.

When using a computer, we can use any single software with drawing capabilities, but some widely used software, such as Microsoft Office's Word, can only create simple graphics and cannot fully meet the requirements for drawing complex DMA Models. Try using a professional graphics software for optimal results.

[10] Although the term "Mind Map" was first popularized by British popular psychology author and television personality Tony Buzan the use of diagrams that visually "map" information using branching and radial maps traces back centuries – Wikipedia

[11] Any pen is good, preferably an erasable pencil, and using different colors can add fun and aid memory.

Here are some options[12] when this book was published (November 2024). Readers could delve into the array of relevant software solutions on the market today for better choices.

Paid Desktop Software:

Visio - A product from Microsoft, widely used in the professional drawing industry.

EdrawMax - A flexible and convenient product from Wondershare, offering a bunch of templates at a reasonable price.

Visual Paradigm - A professional-grade software produced by Visual Paradigm, allows users to create various types of high-quality graphics.

Excel - Spreadsheet software within MS Office suite. It's a good choice for Table-formed Models.

Free Desktop Software:

Draw.io - Open-source with enriched graphics functionalities. Has both desktop and online version.

Visme - Similar to Visio and EdrawMax, the free version has a limit on the number of projects.

Visual Paradigm - Same as the paid version above, students or individuals may get a free license.

Free Online Website: (Advanced features may require a paid subscription)

Google Drawings - Simple interface, no templates, completely free.

Canva - Online comprehensive drawing service, realizes useful functions, and provides many free or paid templates.

Diagrams - like Visio and EdrawMax, allowing users to save drawing files to cloud storage.

Readers can choose any of the software or find better alternatives. Based on my experience, it is advisable to select user-friendly software over price considerations. Free products might be cumbersome to use, resulting in frustration and wasting time on graphics creation and editing. The efficiency of these free products may even be lower than a pen and paper for some situations.

As mentioned earlier, MS Office Word can create basic graphics, but it is not a professional drawing software and has limited efficiency and expandability. Beginners can use it in the initial learning stages and create templates for each DAM model, making it easier to create new model instances once required.

In addition, we can use note-taking software (for personal computers or tablets) that supports electronic pen, mouse or finger to draw on the screen, such as MS OneNote and Sketchbook.

Electronic pens are like physical pen but overcome drawbacks of the latter, such as easier modification and duplications. However, it requires purchasing special hardware that supports screen drawing. People may think of a mobile phone, but because of its small screen size, inconvenient to draw and process the model. Unless for a simple model, not recommended as the primary tool.

For models presented in table format, Microsoft Office's Excel is good enough. Its numerical calculation functions can automatically complete the calculations, such as calculating the total score for options in a Selection & Evaluation Model, the force and relationship strength values for a Things Development & Prediction Model.

Various software programs are constantly being upgraded, and many new software or services are being launched. Please remember, the DMA model is a set of thinking methods universally used, not related to any specific software, and software is only one means to create and present these models.

[12] Personal opinion & experience, no commercial sponsorship.

From the next chapter, readers can learn about DMA models one by one. There is no direct relationship and dependency between these models, so you can jump to any model chapter you are interested in through the model's name in the Table of Content. If you want to learn all Models, it's better to read in chapter order.

For some complex analysis scenarios, we need to create multiple models to work together. In these cases, there naturally exist different sequences and dependency relationships between the models, but this beyond the learning stage. To gain that, you should fully understand all the models before you can apply them skillfully and make sound decisions.

Chapter 3

The Hub of Innovative Thinking - 9−Star Gathering Model

Creativity is intelligence having fun.

Albert Einstein
German/USA physicist [1879-1955]



Innovation is the driving force behind human progress, and individuals who have the ability to think innovatively are often the winners in their careers and personal lives. However, for most people, innovation is not an easy feat to accomplish. The lack of effective tools or methods is the main reason - that's why subjective efforts often fell short and yielded no results.

In this chapter, we present an effective, efficient, and practical model for innovative thinking. By learning this model, you can unleash your innovative potential, let your own innovative ideas spring up and create models that apply to various scenarios, or use new methods to solve old problems, gaining advanced and exceptional returns.

Scenario: The DMA Innovation Model, also called 9−Star Gathering Model can broaden innovative thinking. In the business field, it can provide ideas for creating new products or and transforming/optimizing existed products or services, helping to find new ways to break through and solve problems. It can also be a helpful tool for readers in other fields, such as scientific research, artistic idea/creation, and interpersonal communication, making innovation simple.

It should be noted that when using the same model to address the same issue, individuals may arrive at different outcomes due to their varying personalities, experiences, and backgrounds. Furthermore, innovation itself is a product of unrestricted and free inspiration. However, these differences in outcomes do not impact the objective of using the model, which is to effectively solve the problem at hand.

Analysis purpose should be always the first thing to consider and remember for any model. Although there will be fun creating such innovative models, you need to keep a focus on the purpose of analysis - solve problems and/or gain benefits.

Model Name and Visual Avatars	Benefits of Using this Model
Innovation Model (9−Star Gathering Model) **Visual Avatars:** Shape Resembling: The circular shape comprises nine fan-shaped sections that represent the nine directions for triggering innovative ideas. The central part shows the state/attributes of the subject to be analyzed, serving as a reminder. The nine fan-shaped sections surrounding the central part will be filled with innovative ideas at varying angles. 	• Create Value through Innovative Methods • Solve old problems with new ideas • Create new products and services to outshine competitors • Add inspiration to the creation of artistic works or novel documents • Gain the favor of friends through innovative and engaging methods • Unblock a stuck meeting process • Provide new and impressive training methods for employees

Brief Model Examples

Model Instances	Brief Introduction
Microsoft Refactors the spatial development & deployment of the customer's system.	Many companies have concerns about fully deploying their business systems in the public cloud. To tackle these concerns and explore new revenue growth opportunities, Microsoft launched Azure Stack. This product includes many same services running on-premises that are consistent with Azure Cloud, and also can run in coordination with the latter. This not only captures the market but also lays the foundation to expand its cloud services.
Sony Designs Noise-Canceling Headphones	Sony Corporation has developed a new technology that uses reverse sound waves to counteract out external sounds from headphones. This innovative approach eliminates the need for heavy sound insulation materials, solving the problem of bulky noise-canceling headphones.
Google Develops Cluster Computing Model	Google used computer clusters (multiple computers in multiple groups work together) to achieve a cost-effective increase in computing power (instead of an expensive powerful singular computer) and resolved operational disruptions caused by component failures or shutdowns.
President Nixon Divided and conquered the enemy alliance.	In the early 1970s, President Richard Nixon of the United States pursued a policy of rapprochement with the People's Republic of China. This move was partially aimed at exploiting the Sino-Soviet split and improving relations with China to put pressure on the Soviet Union and potentially weaken the Soviet bloc.

Possible Consequences of Not Using this Model

- Slow product launch & updates beget a disadvantage in the market competition.
- Ongoing issues continue to consume resources and are difficult to eradicate.
- Rigid and conservative, leading to a boring life and unexciting work.

Model Instance Creation Demo (See Section 3.2 for details):

Fire Captain Uses 9−Star Gathering Model to Address California Wildfires.



Protagonist & Story Summary:

• **Sam** is a fire captain in California.

• He and his team have spent several months each year fighting fires on the front line. Despite their exhaustion, the sight of burned homes and grieving people deeply saddened Sam. He contends that implementing long-term, sustainable solutions is crucial to combat wildfires and prevent recurring large-scale disasters.

• The causes and solutions of wildfires are complex, and conventional methods have been implemented for years with limited success. Is Sam able to propose any new innovative solutions to resolve this issue?

• Based on the phenomena he observed during the firefighting process, Sam put forward a variety of new ideas focused on solving problems, not just from the viewpoint of a firefighter.

Expanding Ideas - Self-Exercise & Practice:

Business Breakthrough	Career Development
• Design a completely new product or service • Address Goods Distribution Challenges with Innovative Solutions • Create a new sales model	• Create an impressive professional resume • Discover effective methods to break through career bottlenecks • Integrate a variety of ideas to a team

Competitive Strategy	Social Life:
• Get more support from colleagues • Analyzing and leveraging competitors' weaknesses	• Make a lasting impression on others • Alternative approaches to resolving family conflicts

The DMA Innovation Model can meet the growing demand for innovation, such as creating a new product, finding a new solution to a problem, or using a new method to generate revenue. Visually, it features a circular track divided into nine independent segments, hence named it 9−Star Gathering Model.

Innovation is a hot topic, and many books on innovation have become bestsellers. These books are useful, systematic, and theoretical, but we have not yet seen any Innovation method presented in a visualized model which can reduce the limitations and tedium of information presented in text format, and fully uses the division of labor between the left and right hemispheres of the brain. With the DMA Innovation Model, users can generate various innovative ideas from multiple perspectives in a short amount of time, then evaluate and select the most suitable one.

Innovation has played an extremely important role in human history and is even more necessary in today's rapidly developing economy. Jack Welch, in his book ***Winning,*** mentions that the first strategy for a company's development is to bring in fresh blood. This can be effective, but it also means that some 'traditional' managers and employees are forced to leave, and the existing products or workflows have to undergo significant changes, resulting in definite and substantial investment risk. Injecting fresh blood can certainly break the old thinking pattern and bring in new ideas, but not always the best strategy to cut the

contributions of experienced employees. Although these individuals may pose a hindrance to some uncertain 'change', but their comprehensive knowledge of the product and intricate details is a potential enormous asset to the company.

For company owners, creating an atmosphere and implementing guidelines for innovative thinking does not need expensive, sudden organizational restructuring. Even incremental daily innovation is beneficial. All business owners need is a simple incentive measure and guidelines to foster a culture of innovation within the company. If there is a sense of urgency for innovation, one of the quickest and most cost-effective way for employees and managers at all levels, is to use the Innovation Model introduced in this chapter.

From an employee's perspective, regardless of whether the job is technical or managerial, having a mindset of constant innovation is crucial to avoid being unexpectedly dismissed from a comfortable position. It seems not easy, but learning the Innovation Model can make it simpler. This model is not only engaging and enjoyable, making innovation in the workplace a source of pleasure, but also enhances job efficiency, creating significant additional value for the company and employees themselves.

In creating this model, it is essential to find all directions of innovative thinking, i.e., when creating a new product, combining functions of multiple existing products into one is a direction, referred to one of the **Innovation Direction Pole** (named Aggregation & Connection) in the model. To achieve a comprehensive categorization of innovative models without omissions, it requires a combination of divergent and convergent thinking. Divergent thinking is used for expanding the scope of the thought process, with its flexible and free-flowing nature helping to identify various potential directions for innovation. Convergent thinking, in comparison, is to maintain logical rigor, prevent deviation, and avoid anything important ignored.

The purpose of the above is to provide readers with an understanding of a common thinking process, which involves both divergent and convergent thinking while following a hierarchical and branched thinking pattern of brain and this process is essential in all stages of model design, creation, and utilization.

In the following section, readers can gain insight into the specific introduction of the nine Innovation thinking direction poles in three categories, along with a summary in tabular form with examples.

Section 3.1: Innovation Direction Pole

In the previous section, we briefly mentioned the concept of Innovation Direction Pole (the direction to develop innovative ideas) and provided a simple example of combining multiple product functionalities into a new product. It is certainly to use one innovative direction, but enough. In this section, we will introduce three major categories of innovative directions which group nine specific innovation direction poles. These are the basic components of the Innovation Model and are represented as a fan-shaped structure.

The name and expression of the Innovation Direction Pole may be abstract to readers who are in contact with it for the first time. The following example tables and Model Instance Creation Demo will assist readers in developing a more comprehensive understanding step by step.

Category 1: Internal Composition

Segmentation & Separation: This Innovation Direction Pole separates the various components in the existing object to realize the convenience or new function brought by the change of the composition.

Enlarge & Shrink: Significantly enlarging or reducing the dimensions or scope of the current object (or its components) to go beyond conventional perceptions, creating new functionalities, or addressing problems.

Cluster Expansion: Replicate physical or virtual entities on a large scale to form more powerful groups that support each other to achieve enhanced functions & effects.

Attribute Conversion: Transform certain properties of the current object in the opposite or other direction to achieve desired new properties or convert adversity into advantage.

Category 2: External Relations

Aggregation & Connection: Aggregating external objects or elements into the current object, or connecting internal elements/components to external objects, thus creating previously nonexistent associations between the current object and other objects.

Blockade & Isolation: In contrast to Aggregation & Connection, it is possible to add an external object as a separator/isolator to the current object, disconnecting the current object and another associated object.

The concept of **Blockade & Isolation** is like that of **Segmentation & Separation** above, while the difference lies in the nature of these two categories. Blockade & Isolation refer to the separation between different objects, while Segmentation & Separation refer to the segregation between the internal components of the object.

Cross-Disciplinary Attributes: Extracting the properties of a particular object and applying them to other objects[13].

Category 3: Internal and External Coordination:

Expansion & Extension: Extend or expand the functionality of existing objects by expanding either internally towards other objects or externally towards the object's internal components.

Space & Time Reorganization: Reorganizing the components of a system or rearranging multiple objects in terms of time and space, or reallocating resources to address issues, promote collaborative development to enhance efficiency and productivity.

Innovation Direction Pole (IDP) Examples (Tab-3.1-0)

IDPs	Product Innovation	Solution Innovation	Interpersonal Innovation
Segmentation & Separation	• Detachable Ink Cartridge Fountain Pen • Various wireless devices (physically separated)	• Divide the large, complex, and difficult-to-manage business system into small units based on functions or regions to facilitate management and problem identification. • Employ a strategy to divide enemy coalition, and ultimately defeat each smaller groups one by one.	• Separate the advantages and disadvantages of friends, look more at their advantages to maintain a good relationship. • Separate one's emotions from rationality and approach others with an objective attitude.

[13] This is also effective for learning new knowledge: analyze whether the newly learned knowledge can be applied to other fields, such as business competition, personal career development, product development, etc.

Enlarge & Shrink	• Slide projector • Compressed Biscuits	• Expandable Solar Panels - Increasing Power Supply for Satellites in Space. • Telescopic antenna, to solve radio antenna being too long for convenient portability.	• To establish a close relationship, increase the frequency of communication with friends. • Achieve perfection or exceed expectations in the tasks entrusted by friends and leaders to gain their trust.
Cluster Expansion	• Building Blocks Toys • A communication and positioning network formed by a group of satellites.	• A data center composed of many general-performance computers replaces a single large-scale high-performance computer server. • Multi-warhead missiles with separation technology - disrupting and breakthrough missile interception defense systems. • The 'Three-Three system' tactic developed by the Chinese army (three soldiers form a unit to target a single enemy soldier) can easily destroy enemy personnel during warfare.	• Respect everyone daily, establishing friendly and trustworthy relationships across all levels (security guard and cleaning staff are also worthy of respect). This will facilitate getting help and special attention when needed in the future.
Attribute Conversion	• Solar Panel Street Lights (Converting solar energy into electrical energy) • Innovative Baby Diapers - Upon a baby's urination, the product emits a distinctive odor to alert parents (liquid triggers gas) • Digital cameras use a CCD sensor to convert film (converting light signals from chemical reactions to electrical signals).	• Imitate handwriting, refine the electronic touch screen, and realize pen drawing and writing on the display screen. • The character-based computer operating system has been transformed into a graphical user interface, enabling wider adoption of computer technology.	• When conflicts arise with others, engage in open and honest communication, make some concessions, and turn foes into friends - transforming conflicts into positive relationships. • Transform relationships with classmates and colleagues into friendships.
Aggregation & Connection	• Multifunctional products, such as toothpaste that combines three functions (whitening, cavity protection, and soothing properties) • A multi-interface connector hub for connecting various types of computer devices	• Integrating electrical components onto a printed circuit board(known as a chip), allows for the synergy of multiple functions and the reduction of space utilization. • Joint operations of land, sea, and air forces achieve effects that any single branch of the military cannot achieve.	• Gathering the residents of the community and uniting their efforts can facilitate a prompt resolution of any apartment issues. • In order to attract talent that cannot be sourced through headhunting, extend invitations to them to participate in our company's technical conferences to enhance their impression of

	• Various computer network hardware products		our company. This ultimately leads to successful recruitment.